

Tribhuvan University
Institute of Science and Technology

20812

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Bachelor Level / Third Year /Fifth Semester/Science
Computer Science and Information Technology (CSC326)
(System Analysis and Design)
(NEW COURSE)

Full Marks: 60

Pass Marks: 24

Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Section A

Attempt any TWO questions.

(2 × 10 = 20)

1. List different phases of project management. Explain the different steps involved in the first phase of project management. [2+8]
2. Compare and contrast between observation and document analysis while requirement determination. Explain radical methods for determining the requirements. [4+6]
3. Draw DFD for the following scenario:
A patient after registration in the online pathology system will get login credentials. After the valid login the patient upon entering the details of test to be conducted will get the test date and time. The patient can also pay online. The patient can download the test report from the system after notification. The patient will also get schedule for doctor visit if required and in that case the report will also be emailed to the doctor.

[10]

Section B

Attempt any EIGHT questions.

(8 × 5 = 40)

4. Highlight the critical factors that distinguish agile methods from traditional methods. [5]
5. Illustrate the heart of the software development process. [5]
6. An initial investment of Rs. 5,00,000 is required to start a company. The cash flow-in is Rs. 60,000 per year. Assuming the interest rate as 15%, calculate the discounted payback period. (use Net Present Value) [5]
7. Explain the steps involved in the dialogue design process. [5]
8. Describe the process of designing physical tables. [5]
9. Explain different types of installation. [5]
10. How can maintenance effectiveness be measured? Explain. [5]
11. Draw the sequence diagram for the online food ordering system with the online payment facility. [5]
12. Write short notes on [2×2.5=5]
 - a) Integration testing
 - a) Relational model